

Medicaid Analysis

Organization Introduction

The organization we are working with is Medicaid, a joint federal and state program in the United States that provides free or low-cost health coverage to millions of Americans with limited income and resources. The population it serves is children, pregnant women, seniors, and people with disabilities. Each state runs its own program while following federal guidelines, and there are different benefits and eligibility rules depending on the state. The main difference between Medicaid and Medicare is that Medicaid provides health coverage for people with low incomes, while Medicare provides coverage for those older than 65 and some younger individuals with specific disabilities. Medicaid is jointly funded by the federal and state governments, but the federal government pays states for a specified percentage of program expenditures, and the program serves approximately 83 million Americans.

Medicaid is a great organization to work with since the rules, procedures, and billing codes are governed by federal and state law which are largely set in stone and do not shift based on internal decisions. People supported by Medicaid are not subject to sudden contract renegotiations or policy pivots from a private payer, and Medicaid cannot go bankrupt or be acquired. The demand for Medicaid is consistent since it serves low income and vulnerable populations and that is why making it more cost-efficient is so important.

Problem to Solve

The main problem our group decided to focus on is identifying high cost spending in Medicaid as pharmaceutical costs are increasing very quickly, and that is putting more pressure on healthcare resources. In addition, it is also becoming more difficult to clearly understand what is causing the costs to increase. One of the biggest challenges is identifying the drugs that are driving the highest spending, and some of the most expensive are not prescribed very often so oftentimes they do not immediately stand out right away in the data without cleaning and identifiable analysis. Without clean and well structured datasets, it becomes extremely difficult to figure out which medications are creating the largest financial impact as spending patterns are different across states, due to differing demand and prices of certain types of drugs.

The difference in spending patterns is what makes the problem of identifying high cost areas so difficult and complex. It may also result in differing strategies for combatting high cost spending in different regions of the United States. The problem opens the door to creating cost-efficient spending strategies that Medicaid can implement. Since Medicaid is funded by taxpayers' money, the American people and legislators will certainly be looking to make actionable improvements that certainly make their hard earned money go towards meaningful and worthwhile spending, while still supporting millions of Americans.

Data Sources

The data sources we used for this analysis are pulled from government sources, making it the most reliable source with regards to Medicaid data collection and integrity, as well as population data. The Medicaid dataset is titled *All States 2025*, and includes data that identifies

the drug name, national drug code (NDC), number of prescriptions, reimbursement amounts, state, and reporting period. The dataset allowed us to identify high cost drugs, utilization patterns, state-level reimbursement differences, and potential areas for savings. The source allows us to analyze and identify these key factors across the United States, and provides a picture of the reality of drug spending. The second data source we used is 2024 population data from the census government website. This data source is also the most reliable data we could obtain regarding population figures for the 50 US states we need to analyze Medicaid data per capita. The census population data allowed us to better understand Medicaid reimbursement and spending relative to state size, extremely necessary as we know that size and spending have a strong positive correlation.

Analytical Techniques

To conduct a detailed analysis of Medicaid pharmaceutical spending, our group applied several analytical techniques such as descriptive analytics, comparative analysis and exploratory data analysis. We began by preparing a copy of the dataset so that we get a clear ground to work on. In our previous iterations of the project, we used python to conduct the analysis, but after receiving feedback from the second iteration, we decided to only use Tableau. This decision was made because we realized that our group was more comfortable and knowledgeable in working with Tableau software than writing python code. We began by loading the dataset into Tableau and the over 5 million records of data took quite some time to fully load as a data source in the desktop platform. Then we began by cleaning up data types to better fit the analysis needed. The fields NDC, Labeler Code, Product Code, Package Size, Year, and Quarter were all changed from number to string data types because they were not necessary to aggregate, nor needed for any other purpose other than filter by, or group by cases. Specifically the Year and Quarter fields were used to create the calculated field Date [defined by `MAKEDATE(INT([Year]), (INT([Quarter])*3)-2, 1)`], which forces the fields to be integers and then transforms them into dates corresponding to the first day of each quarter. We did this in case we wanted to analyze trends over time, however we decided against that analysis since the data is 2025 only and that is not sufficient data to accurately represent trends over time.

The next task was joining the population data to the Medicaid data in Tableau. I initially downloaded the population data and then worked in the Excel file to make it compatible with the Medicaid data. The Medicaid data included States by the two-letter abbreviation "AZ" for example. The 2024 population spelled out the name of the state, so my first task was to create a separate column and input the two-letter abbreviation per state. I then needed to delete population figures for years 2020 - 2023 because that data is stale and not needed for our analysis, in addition to deleting a few very top and very bottom rows that contained information related to the context of the data, not the actual data itself. The population data was then ready to be loaded into Tableau and the full outer join occurs on the Medicaid data State field and the census data State Abbreviation field. The next step was to continue making the calculated fields which include Medicaid Reimbursement per Capita [defined by $\text{SUM}([\text{Medicaid Amount Reimbursed}]) / \text{SUM}([2024 \text{ Population}])$], Non-Medicaid Reimbursement per Capita [defined by $\text{SUM}([\text{Non Medicaid Amount Reimbursed}]) / \text{SUM}([2024 \text{ Population}])$], and Reimbursement per Capita [defined by $\text{SUM}([\text{Total Amount Reimbursed}]) / \text{SUM}([2024 \text{ Population}])$]. The calculated fields are the necessary data we need to accurately analyze and evaluate Medicaid spending.

Client Discussion and Scope Refinement

Because our project used public Medicaid and census data, rather than a direct client provided dataset, our scope was refined through instructor feedback and group discussion instead of traditional client meetings. After the second iteration, we narrowed the project away from a python heavy analysis, toward a Tableau based approach that better analyzed and communicated the findings visually. The difficulty associated with this project scope is the scale of Medicaid as it is the largest medical provider in the nation and understanding its impact and policies within a project of this scale is difficult. We decided to focus on two narrow opportunities of savings. The final scope was more practical because it emphasized a narrow approach where Medicaid could identify the highest reimbursement pressure and where targeted intervention may create the greatest financial impact.

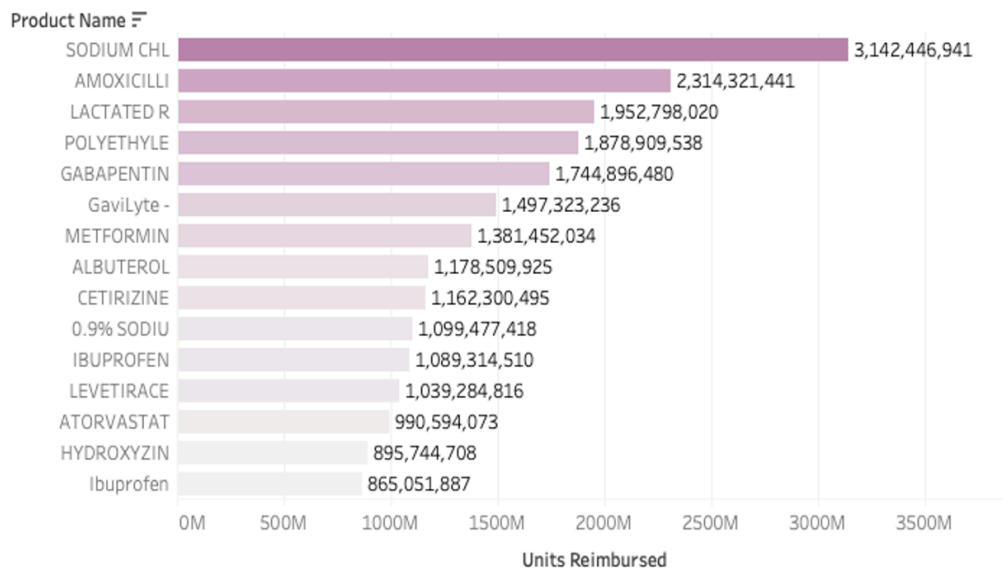
Challenges Faced

Throughout the course we faced many challenges but the most important was communication and collaborating together as we had some small issues related to our mode of communication. We tried using discord but later decided on messages as it was difficult to get everyone on discord at the same time, due to other responsibilities and timing clashes. We ended up using discord once a week to meet and talk about the upcoming deadlines, in addition to any ideas we had related to project planning. We made adjustments and this significantly improved our communication, but during critical times like deadlines, we found each other difficult to reach and largely lacking responsibility for actions and completing work on time. We regrettably found ourselves unable to solve the issue through the last submission of the project.

Another problem we faced as a group was from differing perspectives regarding the overall direction and structure of the project as group members were not entirely aligned on the focus of our topic and our analytical approach. We worked through these differences through lengthy discussions on discord and were successful in establishing a unified project direction. The final main issue we faced as a team was selecting the most appropriate analytical tools for our analytics. In the first and second iteration, we used python to analyse and clean our datasets, but after receiving feedback after the second iteration, we switched to Tableau and were successful in meeting the goal of our project.

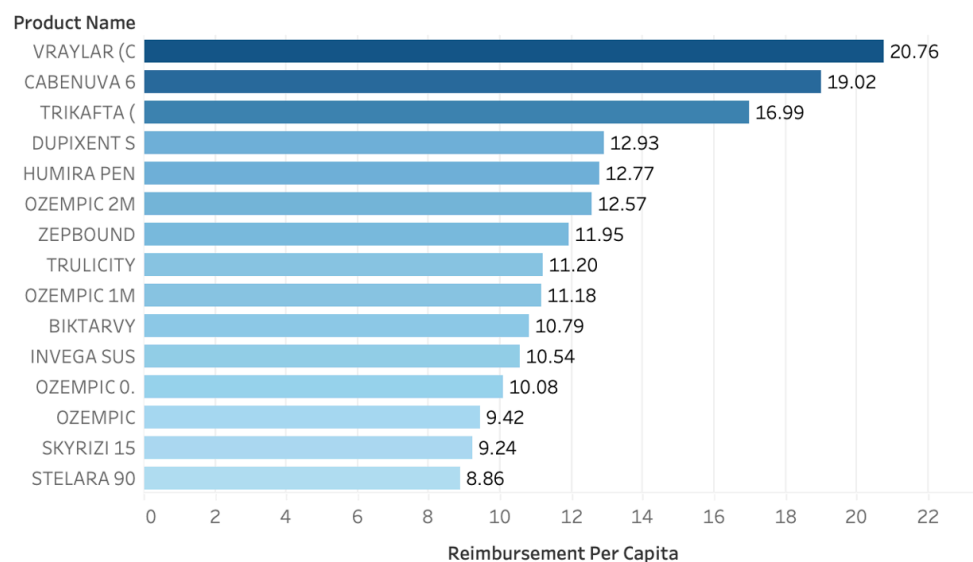
Visualization Deliverables

Top 15 Product Units Reimbursed



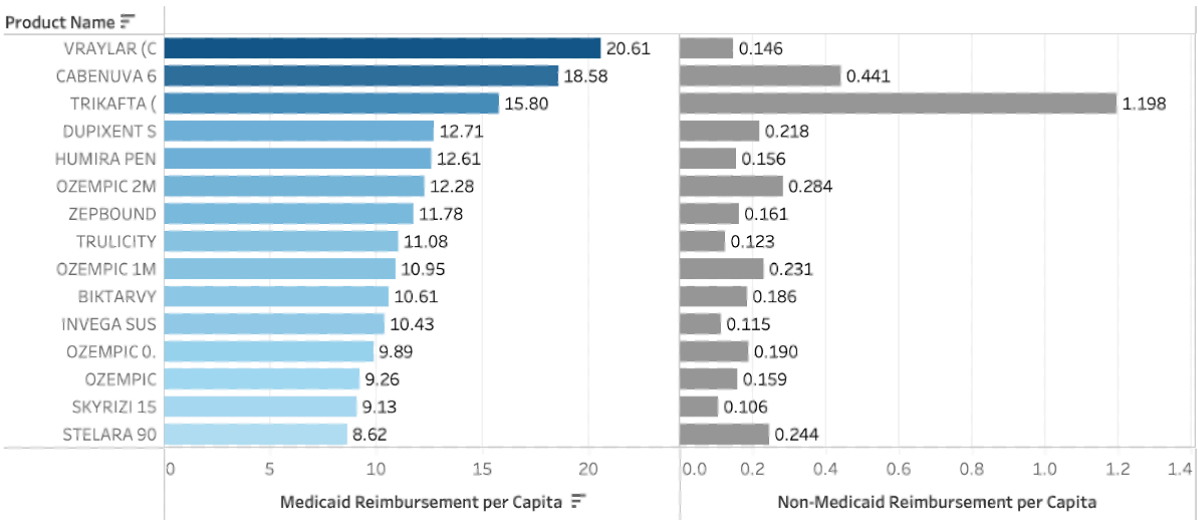
This analysis shows the top 15 product units reimbursed as a horizontal bar chart. There was no calculated field needed for this analysis, and the color scheme allows us to see the degree of descent over the highest volume drugs Medicaid reimburses.

Top 15 Drugs per Capita



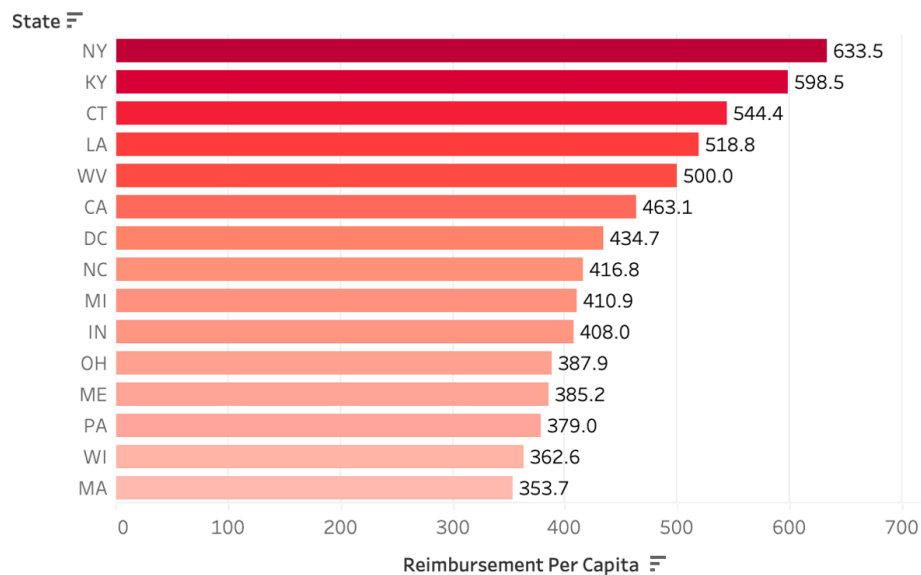
This analysis shows the top 15 high-cost drugs per capita. The calculated field “Reimbursement per Capita” was used to identify the high-cost drugs.

Top 15 Drugs Medicaid vs Non-Medicaid Reimbursement per Capita

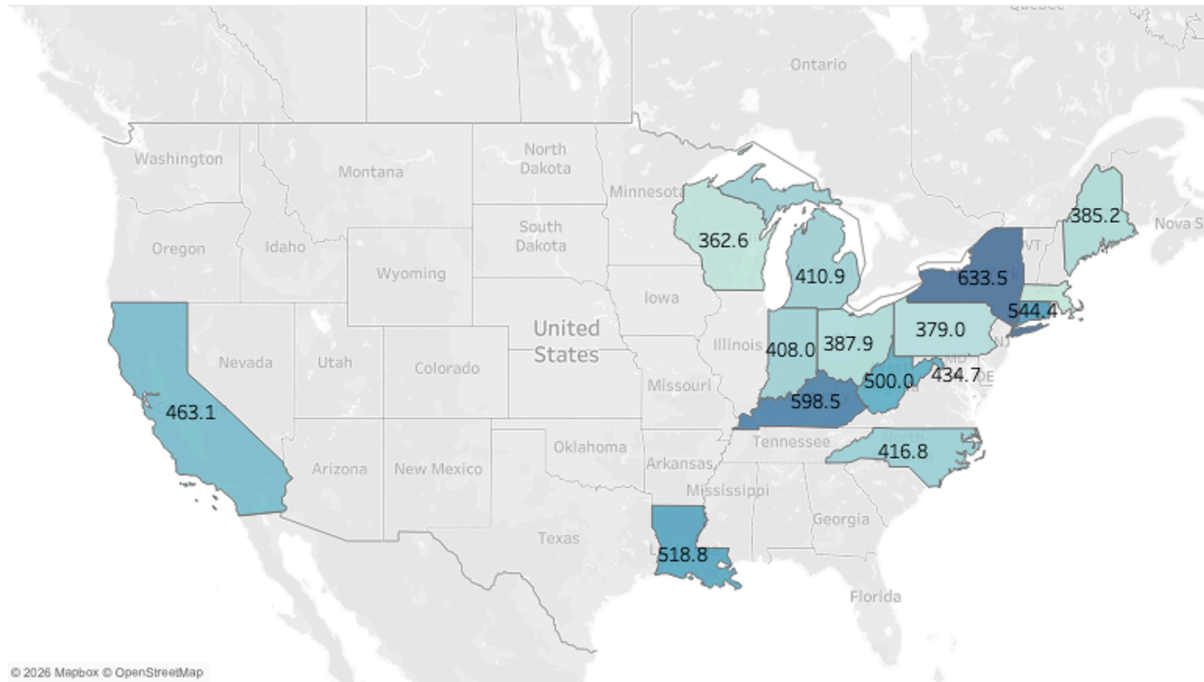


This analysis shows the top 15 drugs Medicaid vs Non-Medicaid reimbursement per capita in a side-by-side horizontal bar chart. The fields “Medicaid Reimbursement per Capita” and “Non-Medicaid Reimbursement per Capita” were used here to show the cost sharing responsibilities for these drugs.

Top 15 States Spending per Capita



This top 15 states spending per capita analysis was created using the “Reimbursement per Capita” calculated field, and identifies the highest spending states relative to their population. It is helpful when determining if state-level policy intervention is necessary.



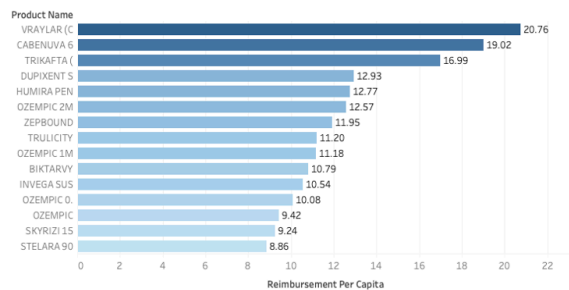
The geographic mapping of state-level spending per capita identifies that there is a concentration in the northeast sector, as well as some southern outliers, where Medicaid spending is concentrated. This helps identify if there are regional demographic or policy implications, and the potential for regional policy intervention for cost savings.

Top 15 States Trend Spending per Population

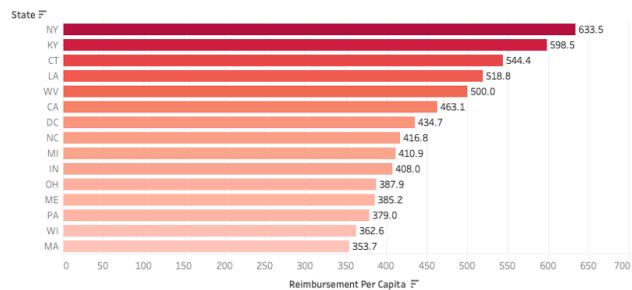


This analysis of the top 15 highest total amount Medicaid reimbursement mapped against state population shows that population size has a strong positive correlation with total amount reimbursed. This supports analyzing the cost per capita over general total reimbursement for the greatest impact cost-savings opportunities.

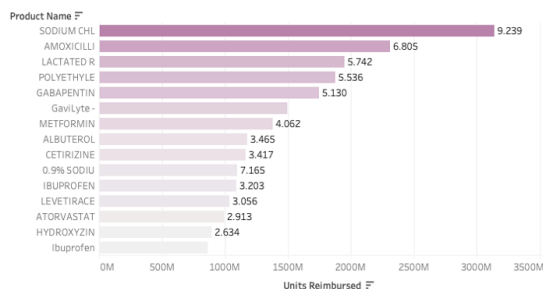
Top 15 Drugs per Capita



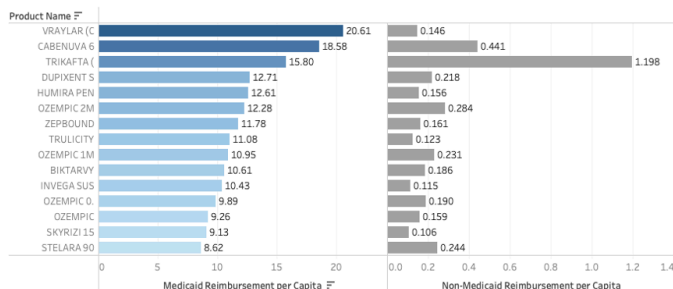
Top 15 States Spending per Capita



Top 15 Product Units Reimbursed



Top 15 Drugs Medicaid vs Non-Medicaid Reimbursement per Capita



This is an example of an interactive dashboard that Medicaid analysts can use to actively look for ways to be more cost-efficient.

Results of Analysis

The analysis showed that the largest portion of Medicaid spending is not where people initially think it is. Initially, we started the analysis at the drug-level to see where the cost concentration lies in that sphere of Medicaid spending. The analysis “Top 15 Product Units Reimbursed” showed that the highest volume drugs are everyday medications that are IV fluids, antibiotics, diabetes, and allergy drugs. This is reassuring from a cost-saving perspective because it tells us that the high prescription volume drugs are not the problem. The drugs that Medicaid reimburses the most are low-cost generic drugs, so then the real cost pressure must come from somewhere else. The analysis “Top 15 Drugs per Capita” shows that specialty drugs like Vraylar, Cabenuva, and Trikafta are the leading high-cost drugs. Those three drugs treat specific illnesses schizophrenia, HIV, and cystic fibrosis, and there is no generic alternative for such treatment. This analysis reinforces that the high-cost specialty drugs are driving costs, not the highest volume drugs like most people would assume. The analysis “Top 15 Drugs Medicaid vs Non-Medicaid Reimbursement per Capita” builds upon the previous analysis that identifies the high-cost drugs. This shows the breakdown between Medicaid and non-Medicaid reimbursement for the top 15 high-cost drugs, and we can see that Medicaid is carrying nearly

the entire financial burden for every drug on the list. The non-Medicaid reimbursement is extremely minimal and Trikafta is the only outlier in this case, meaning that there is very little cost sharing happening. This reaffirms that Medicaid needs to target the high-cost drugs through exploring rebate negotiations or supplemental rebate agreements.

The next task was to look at the state-level Medicaid spending, and identify if there are states that have a high reimbursement per capita value while being a lower population state. The analysis “Top 15 States Spending per Capita” identifies New York, Kentucky, Connecticut, Louisiana, and West Virginia with the highest spending per capita. New York is relatively understandable as they have a massive population and varying demographics, however the other four states are not anywhere close to high population in comparison to other US states. The per capita normalization reveals that those smaller states are carrying disproportionately high-costs relative to their populations. This could be explained if these states have higher rates of chronic illness and/or Medicaid enrollment, but this also makes them priorities for state-level cost management strategies and enhanced controls. The data from that analysis can be mapped geographically to see if there is a concentration, which we can tell from the map that there is in the northeast region of the US with few outliers in the south as well. This indicates that there may be regional patterns or state-level policies that drive their cost and can also be explained by shared economic, demographic, or policy characteristics. The current strategies at the state-level aspect of Medicaid can be more effectively approached regionally and it may be more effective than treating each state in isolation since they likely have similar policies and demographics of population that can be addressed at once. The analysis “Top 15 States Trend Spending per Population” is a scatter plot with a trend line to confirm that there is a correlation between population size and total Medicaid reimbursement amount. The R^2 value of 0.95377 and $p < 0.0001$ indicates that there is a strong positive correlation between the two, and our analysis of per capita reimbursement amounts is a more accurate representation of true cost.

Implications of Results

After synthesizing the results from our Tableau visualizations, the main implication we got is that Medicaid should focus its cost control efforts on the areas that are creating the greatest financial pressure instead of treating all pharmaceutical spending the same. The analysis shows that the strongest opportunities for savings are concentrated in a handful of high cost drugs and high spending states. First, Medicaid should prioritize rebate negotiation and prior authorization for the top high cost drugs such as Vraylar, Cabenuva, and Trikafta that create a much higher reimbursement burden than many commonly reimbursed products. Medicaid should prioritize for stronger rebate negotiation and review therapeutic alternatives where possible, and apply prior authorization where appropriate as this approach would allow Medicaid to focus on the drugs creating the greatest reimbursement burden while still protecting access to medically necessary treatment. Second, Medicaid should target drug cost rather than prescription volume alone as the results showed that many high volume products are common and low cost medications which means prescription count by itself is not the best indicator of financial impact. A better approach is to focus on reimbursement impact, especially drugs with high reimbursement per capita as a drug may appear frequently in the dataset, but that does not automatically mean it is creating the largest financial burden.

The Medicaid versus non Medicaid reimbursement analysis also suggests that Medicaid bears nearly all of the cost for many of the top high cost drugs, with very limited cost sharing from non Medicaid sources which can be combated by having stronger supplemental rebate agreements and mixing strategies for the payers according to the circumstances. The state-level analysis shows that New York, Kentucky, Connecticut, Louisiana, and West Virginia are the top five highest Medicaid spenders per capita in the nation, which is surprising as smaller states such as Kentucky, West Virginia, and Louisiana are carrying costs far out of proportion to their population size. Overall, the implication of our analysis is that Medicaid can reduce cost pressure more effectively through targeted, data driven strategies, rather than broad, general spending reduction strategies.

Conclusion

In conclusion, our analysis reaffirms that Medicaid's long term sustainability is primarily dependent on its ability to strategically address and manage the key factors driving pharmaceutical cost growth. As Medicaid drug spending continues to increase with the expansion of higher cost specialty medications, data-driven analysis becomes more essential for implementing cost reduction strategies and policies that maintain access to care while improving financial efficiency. Our analysis findings confirmed that Medicaid spending is heavily concentrated among a limited number of high cost drugs and disproportionately high spending states, presenting clear opportunities to focus intervention strategies and produce a significant financial impact. By prioritizing targeted drug reviews, strengthened rebate negotiations, prior authorization strategies, and state level policy reform, Medicaid can implement more effective cost management solutions than broad spending reductions alone. Ultimately, early intervention and data informed policy development will serve as essential tools for improving financial sustainability and maximizing the efficiency of public funding while preserving reliable access to medical coverage for the millions of Americans who rely on Medicaid.